

Today's Special

- Prime Numbers

- Quizzes

- Sqrt

- linked in influencer level speeches

- Next 2 Months topics & schedule

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(Amazon)



2 Months

N, 1 & itself are factors, won't hold if $N=1$

① - factors: 1, Not a prime

$N \geq 1$, No. of factors = 2

```
bool isPrime(N) {
    cnt = 0;
    for (i = 1; i <= N; i++) {
        if (N % i == 0) {
            cnt++;
        }
    }
    if (cnt == 2) {
        return True;
    }
    else return False;
}
```

Assumption:

10^8 iterations in 1 sec

Input:	Iterations	Execution Time
<u>$N = 10^9$</u>	10^9 iterat	10 sec
<u>$N = 10^{18}$</u>	10^{18} iteration	<u>10^{10} sec</u>

10^8 iterat - 1 sec
 10^{18} iterat - n

$$n \times 10^8 = 1 \times 10^{18}$$

$$n = \frac{10^{18}}{10^8} = 10^{18-8} = 10^{10} \text{ sec}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

10^{10} sec $\rightarrow$ hours $\rightarrow$ days $\rightarrow$ Years: 317 Years

1)

$$a \times b = N, b = N/a$$

Both (a, b) factors of N

$(a, N/a)$ factor of N

If a is factor of N
 N/a is also a factor

2) Relational operators

$<, >, <=, >=$

$=, !=$

3) $a \times = 10$

Man $a = 10$

$a \times = N$

Man $a = N$

$$N = 24$$

<u>i</u>	<u>N/i</u>
1	24
2	12
3	8
4	6
6	4
8	3
12	2
24	1

Part-I (rows 1-4)
 Part-II (rows 6-24)

$$N = 100$$

<u>i</u>	<u>N/i</u>
1	100
2	50
4	25
5	20
10	10
20	5
25	4
50	2
100	1

Part I (rows 1-10)
 Part II (rows 20-100)

To get i
 $i = N/i$
 $i^2 = N$
 Sqrt on both
 $i = \sqrt{N}$

obs1: If we iterate on Part I we can get all factors of N .

obs2: In part I $i = 1$

obs3: In part I max $i = \sqrt{N}$

$$\left\{ \begin{array}{l} [1 \sqrt{N}] \\ [1 \sqrt{N}] \end{array} \right\}$$

If N is prime

$[1 \sqrt{N}]$, how many factors = 2

$[1 \sqrt{N}]$, how many factors = 1

$[2 \sqrt{N}]$, how many factors = 0

$$[1 \ 10] = 10$$

$$[3 \ 14] = 14 - 3 + 1 = 12$$

$$[a \ b] = b - a + 1$$

for input $N \Rightarrow [2, \sqrt{N}]$

Iteration $\Rightarrow \sqrt{N} - 1 \approx \sqrt{N}$

isPrime (N) if $i * i \leq N$

```

bool ch = True;
if (N == 1) return False;
i = 2; (i * i <= N); i++;
    if (N % i == 0) {
        ch = False; break;
    }
return ch;

```

Input N	Iterations	Result
10^{18}	10^9	1054

317 years $\rightarrow$ 1054

1 sec = 10^8 iterations / s

$$\sqrt{10^8} = 10^4 = 10,000 \approx 10$$

→
208)

Sum of first 100 Natural Numbers = ? $\frac{50(100)(101)}{2} = 5050$

$$\begin{array}{ccccccccccc} S = & 1 & + & 2 & + & 3 & + & 4 & + & 5 & + & \dots & + & 98 & + & 99 & + & 100 \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \downarrow \\ S = & 100 & & 99 & & 98 & & 97 & & 96 & & & & 3 & & 2 & & 1 \end{array}$$

$$2S = 101 + 101 + 101 + 101 + 101 + \dots + 101 + 101 + 101$$

$$2S = \frac{(100)(101)}{2}, \quad S = \frac{(100)(101)}{2}, \quad S = \frac{50(101)}{1}$$

† Gauss

Sum of first N Natural Numbers = $\frac{(N)(N+1)}{2}$

$$\begin{array}{ccccccccccc} S = & 1 & & 2 & & 3 & & 4 & & 5 & & \dots & & N-2 & & N-1 & & N \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \downarrow \\ S = & N & & N-1 & & N-2 & & N-3 & & N-4 & & & & 3 & & 2 & & 1 \end{array}$$

$$2S = (N+1) + (N+1) + (N+1) + (N+1) + (N+1) + \dots + (N+1) + (N+1) + (N+1)$$

$$2S = (N)(N+1)$$

$$\left\{ S = \frac{(N)(N+1)}{2} \right\}$$

Quiz: 5 10:15 PM] $15/2 = 7$ | $25/2 = 12$

Given N , how many times we need to divide by 2 to

make it 1. $\approx \log_2 N$

$(2) \rightarrow 1$

$1 = \log_2 2^1$

$(4) \rightarrow 2 \rightarrow 1$: $2 = \log_2 2^2$

$2 = \log_2 2^2$

$(8) \rightarrow 4 \rightarrow 2 \rightarrow 1$: $3 = \log_2 2^3$

$3 = \log_2 2^3$

$(9) \rightarrow 4 \rightarrow 2 \rightarrow 1$: $3 = \log_2 9$

$3 = \log_2 9$

$10 \rightarrow 5 \rightarrow 2 \rightarrow 1$: 3

$15 \rightarrow 7 \rightarrow 3 \rightarrow 1$: 3 = $\log_2 15$

$27 \rightarrow 13 \rightarrow 6 \rightarrow 3 \rightarrow 1$: 4 = $\log_2 27$

$\log_a a^N = N$ ✓

$\log_2 2^5 = 5$

$\log_2 2^N = N$

$\log_2 9 = 3.16125$

$\log_2 27 = 4.7548$

$\log_2 15 = 3.9068$

Perfect Square:

N →

1000: { Please learn }
What's $\log$ }

Perfect Square (Amazon)

Given $\{N(\text{Perfect Square})\}$

find its Sqrt $N > 0$

$$N = 100 \rightarrow 10$$

$$N = 25 \rightarrow 5$$

$$N = 20 \rightarrow$$

Sqrt(N) {
 $[1, \sqrt{N}] : \sqrt{N}$
 $i = 1; i \leq N; i++$ {
 if ($i * i == N$) {
 return i;
 }
 }
}

a) N	$N = 100$ $i = 10, 10 * 10 = 100, \text{return } 10$
b) $N/2$	$N = 36$ $i = 6, 6 * 6 = 36, \text{return } 6$
c) $\sqrt{N}$	$N = 64$ $i = 8, 8 * 8 = 64, \text{return } 8$
d) $\log N$	

$$N=100 \text{ } \{ \equiv \}$$

center:

$$[1 \quad \downarrow \quad 100] : 100 \quad \underline{50} \quad \overset{x}{50} \cdot 50 > 100 \quad [51, 52, 53, \dots, 100]$$

$$[1 \quad \underline{49}] : 49 \quad \overset{x}{24} \quad 24 \cdot 24 > 100 \quad [25, 26, 27, \dots]$$

$$[1 \quad \underline{23}] : 23 \quad \underline{11} \quad 11 \cdot 11 > 100 \quad [12, 13, 14, \dots]$$

$$[1 \quad 10] : 10 \quad \underline{5} \quad \overset{x}{5} \cdot 5 < 100 \quad [\overset{x}{4}, \overset{x}{3}, \overset{x}{2}, \overset{x}{1}]$$

$$[6 \quad 10] : 5 \quad \underline{8} \quad \overset{x}{8} \cdot 8 < 100 \quad [\overset{x}{7}, \overset{x}{6}]$$

$$[9 \quad 10] : 2 \quad \underline{9} \quad \overset{x}{9} \cdot 9 < 100 \quad [\overset{x}{7}]$$

$$[10 \quad 10] : 1 \quad \underline{10} \quad \overset{x}{10} \cdot 10 = 100$$

$$[1 \quad \overset{\downarrow}{\quad} \quad N]$$

$N/2 \quad N/2$

Iterations

$$N \rightarrow N/2 \rightarrow N/4 \rightarrow N/8 \rightarrow \dots \approx 1 \rightarrow \log_2 N$$

$$\sqrt{N}, \log_2^N \quad \left. \vphantom{\sqrt{N}} \right\} N = 2^{60}$$

$$\begin{array}{l|l} \sqrt{2^{60}} & \log_2^{60} \\ \hline = 2^{30} & 60 \\ \hline \approx \underline{10^9} & \Rightarrow 60 \end{array}$$

Undo/Redo } Stack
← →

Mo ney } Autocomplete
bile } Trie
onday }

1) knowing —

2) Where to use

→ Tasks / Interview

Intermediate Batch

1) Bit Manipulations : 2

2) Time Complexity : 2

3) Arrays : 8 sessions

4) Hashing / Strings : 4

5) Recursion : 3

6) Subsets / Subsequences : 2 sessions

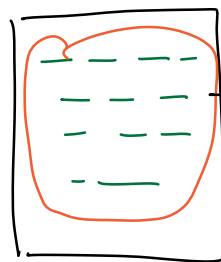
7) Ques : 2/3

Facebook : Mutual Friends

BFS (breadth)

Google Maps : Dijkstra's Algo

Word Pad



Dynamic Program
anning

→ language Independent

→ Tues / Thurs / Satur

→ 9 PM

9 PM - 10:20 PM

10:20 PM - 10:30 break

10:30 PM - 11:30 PM

11:30 PM - 12 AM (Doubt)

→ PDF, with everyone (Dashboard)

→ ≈ 6 assignment Problems

→ Weekly ≈ 20

→ Homework

→ 8/9 Wkly ≈ 160-180 ≈ 150 Classwork

→ Assignment → Doubts

1) Self Doubts

TA

(Teaching Assistant)

Ask Me

Session Doubt/Concept

↓

1) Ask Session

2) Doubts Session

3) Ping me in Slack

Doubts: }